



YISHUN INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAM
Higher 2

NAME

ANSWERS

INDEX NO

CG

BIOLOGY

9744/02

Paper 2 Structured Questions

31 Aug 2022

Candidates answer on the Question Paper.

2 hours

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, index no. and CG on this cover page.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer all questions in the spaces provided on the Question paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question

At the end of the examination, **submit booklets A, B and C separately** to the invigilator.

For Examiner's Use	
Section A	
1	12
2	9
3	10
4	7
5	9
6	9
7	9
8	11
9	12
10	7
11	5
Total	100

This document consists of **29** printed pages and **1** blank page.

Answer **all** questions

- 1 Collagen is a major component of the cartilage found in some of the structures of the human gas exchange system. Cells that synthesise and secrete the components of cartilage are known as chondrocytes.

Fig. 1.1 is a transmission electron micrograph of a chondrocyte.

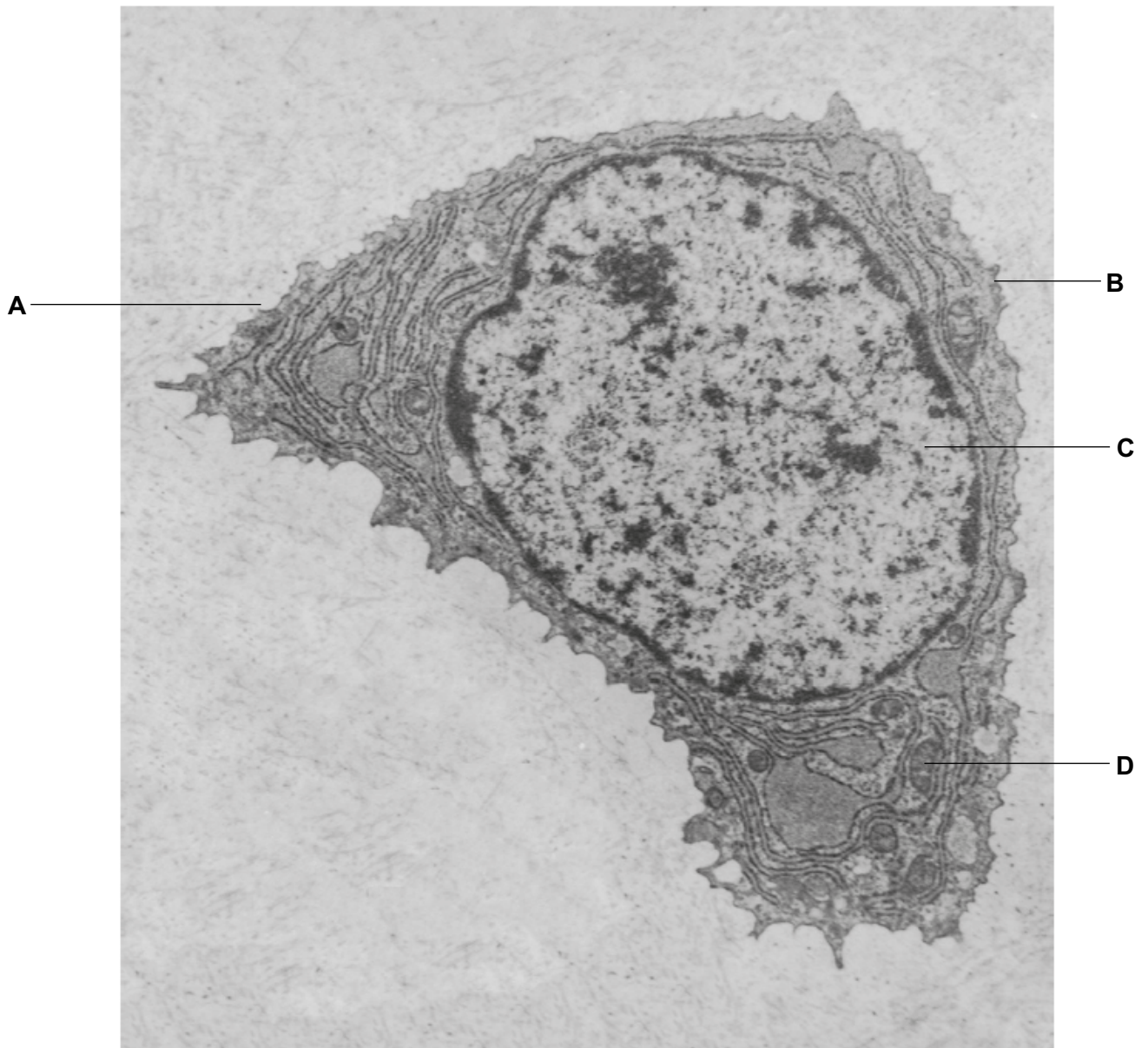


Fig. 1.1

(a) Identify structures A - D.

A rough endoplasmic reticulum;
 ® *rER*
 Ⓐ ribosomes

C nucleus / nucleoplasm /
 euchromatin;
 ® *nucleolus*

B cell surface membrane;

D Mitochondrion;
 ® *mitochondria*

[2]

- (b) With reference to Fig. 1.1, explain **two** features of the chondrocyte that show how the cell is adapted to its function.

1. extensive rough endoplasmic reticulum

with bound ribosomes for synthesis of collagen / components of cartilage

® *protein synthesis (too vague)*;

2. many mitochondria to produce ATP

needed for transcription / translation / collagen synthesis / cartilage component;

3. large nucleus

active transcription of genes coding for collagen;

[2]

any 2 marking points

- (c) Collagen forms tropocollagen which is transported to the Golgi apparatus before being secreted out of the chondrocyte.

Describe the role of Golgi apparatus in collagen formation.

1. receive tropocollagen in vesicle at cis face

tropocollagen undergo sorting, packaging and modification / glycosylation / cleavage of terminal ends;

2. to form molecular identification tag

pinch off in secretory vesicle at trans face;

[2]

(d) Fig. 1.2 shows part of the primary structure of a collagen polypeptide.



Fig. 1.2

With reference to Fig. 1.2, name the

(i) type of covalent bond formed between the amino acids;
peptide bond; [1]

(ii) enzyme that catalyses the formation of bond between the amino acids;
peptidyl transferase; [1]

(iii) reaction that forms the covalent bond between the amino acids.
condensation; [1]

(iv) Fig. 1.3 shows the molecular structure of the amino acid glycine (gly).

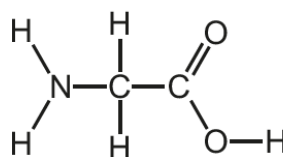


Fig. 1.3

With reference to Fig. 1.2 and Fig. 1.3 and the function of collagen, explain how the structure of a collagen polypeptide makes it suitable to form a collagen molecule.

1. collagen polypeptide has gly-x-y motif /

every 3rd amino acid being gly / 1 in 3 amino acid is gly;

2. gly is smallest amino acid / has H as R-group

fit into tight center of triple helix / tropocollagen allowing it to coil tightly;

3. ref. to NH of peptide bond of gly can form hydrogen bond

with C=O of peptide bond of amino acid in adjacent polypeptide chain;

4. linking any structural feature to conferring higher tensile strength; [3]

any 3 Marking points

[Total: 12]

- 2 (a) The content of lysosomes in animal cells has a pH of 5.0. The cytosol has a pH of 7.2. The lysosomal membrane enclosing the lysosomal content, controls the passage of hydrogen ions from the cytosol into the lysosome. The low pH created by the entry of hydrogen ions is optimum for the action of acid hydrolase enzymes in the lysosome.

- (i) State and explain the transport mechanism used to move hydrogen ions from the cytosol of an animal cell into the lysosome.

Transport mechanism **1. Active transport;**

Explanation **2. Lysosome has a higher concentration of hydrogen ions compared to the cytosol;**

3. ATP is needed to transport hydrogen ions

against a concentration gradient from cytosol into lysosome; [3]

- (ii) Suggest how the structure of the lysosomal membrane allows hydrogen ions to be transported into the lysosome but does not allow the ions to leave the lysosome.

Any 2 below:

1. Hydrogen ions are charged,

∴ will be repelled by the fatty acid tails in the hydrophobic core of the lysosomal membrane (∴ unable to leave the lysosome);

2. Movement of hydrogen ions into the lysosome require a transmembrane protein

embedded in the lysosomal membrane;

3. Transport protein for hydrogen ions is unidirectional,

only transports ions into the lysosome;

4. Binding site for hydrogen ions located on cytosol side

of transmembrane protein in lysosomal membrane; [2]

- (iii) The acid hydrolases in the lysosome cannot function in neutral conditions (pH 7.0) or alkaline conditions.

Explain the advantage to the animal cell of having acid hydrolases that cannot function in neutral, near neutral or alkaline conditions.

1. Acid hydrolases break down / hydrolyse named macromolecule e.g.

proteins, nucleic acids, lipids, and carbohydrates;

2. Acid hydrolases will be denatured at pH of cytosol;

OR

3. Hence inability to function at neutral / alkaline conditions ensures that

.....
damage / break down of cell contents / organelles / molecules in the cytosol is avoided;

[2]

- (b) An analysis of a membrane protein located on the lysosomal membrane is carried out. The amino acid sequence that makes up the **transmembrane** segment of the protein is shown below.

...Ile – Thr – Leu – Ile – Tyr – Phe – Gly – Val – Met – Ala – Gly – Val – Ile – Gly – Thr – Ile – Leu – Leu – Ile – Ser-...

Suggest why such an amino acid sequence would enable the protein to span the lysosomal membrane.

1. R groups of amino acids making up transmembrane segment of protein

.....
are largely non-polar / hydrophobic;

2. Amino acid sequence / polypeptide is folded such that transmembrane region embedded in membrane contains amino acids with hydrophobic R groups facing protein exterior

.....
will interact with non-polar fatty acid tails of phospholipids via hydrophobic interaction;

[2]

.....
 [Total: 9]

- 3 Cyclooxygenase (COX) is an enzyme that catalyses the conversion of arachidonic acid (AA), into prostaglandins which are then involved in promoting inflammation, pain and fever.

There are two isoforms of the enzyme, COX-1 and COX-2 and they differ in the shape of their active sites.

Fig. 3.1 shows a simplified diagram of the active sites in COX-1 and COX-2.

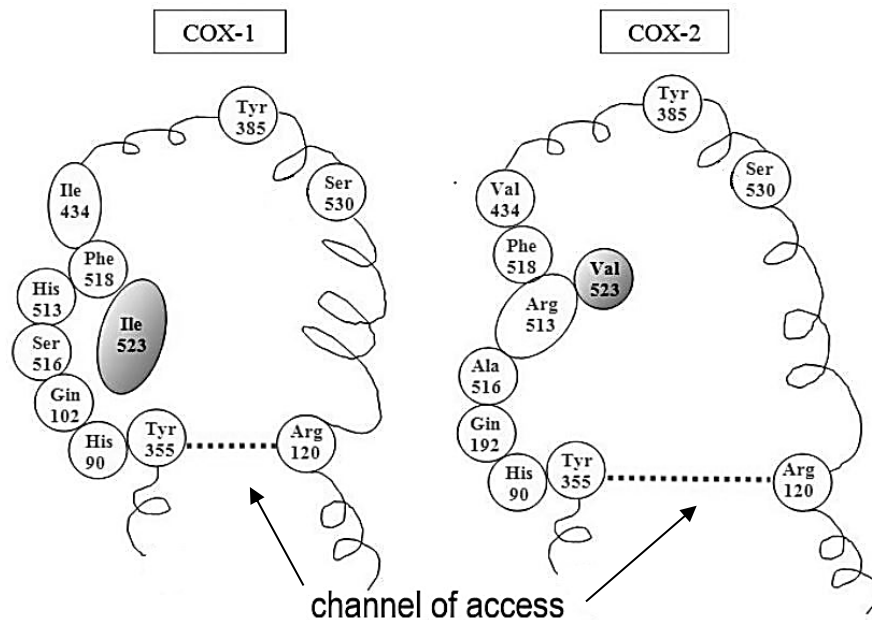


Fig. 3.1

- (a) With reference to Fig. 3.1, describe how the differences in amino acids result in a change in the configuration of the active site in COX-1 and COX-2.

1. QV any differences in amino acids with positions / QV any 2 differences in amino acids without positions

e.g. Gln in COX-1, at position 102 while in COX-2, at position 192 / His is a smaller amino acids in position 513 in COX-1 while arg is a bigger amino acid in same position in COX-2 etc;

2. Different amino acids,

different R groups;

3. Changes in the types of R group interaction between amino acids

Affected folding of polypeptide into its tertiary structure;

4. Resulted in a larger channel of access in COX-2

and a smaller channel of access in COX-1;

[4]

COX enzymes are targets of nonsteroidal anti-inflammatory drugs (NSAIDs) e.g., Ibuprofen, to reduce the inflammatory symptoms.

Fig. 3.2 shows COX-1 when bound to AA and when ibuprofen is present.

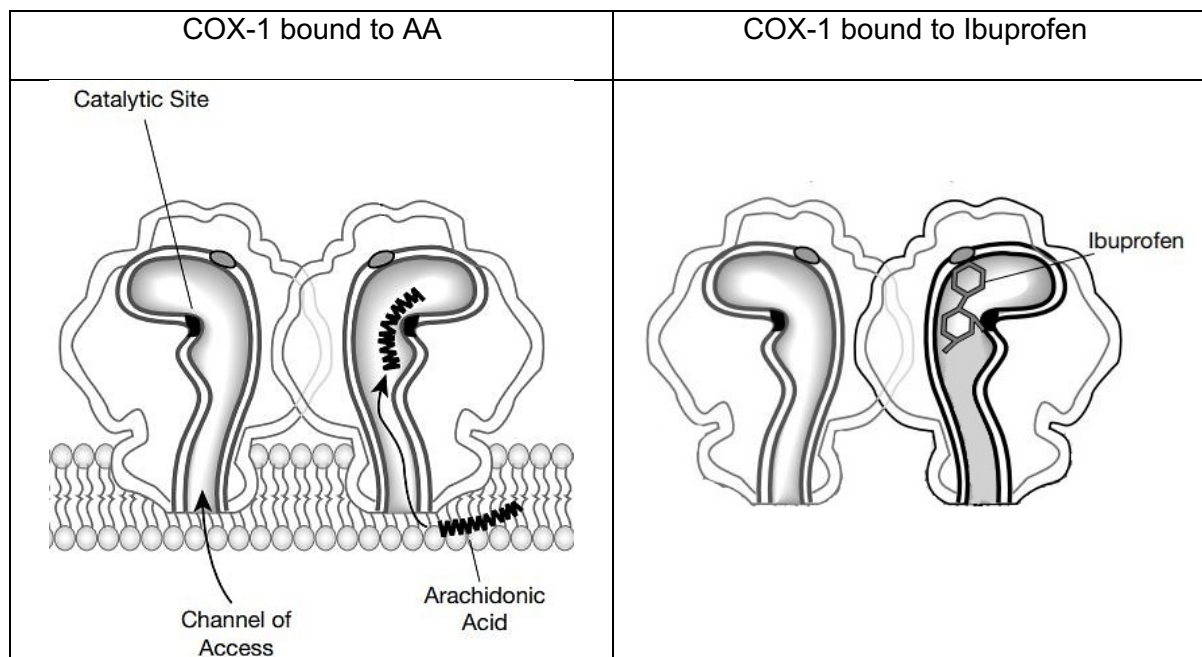


Fig. 3.2

(b) With reference to Fig. 3.2,

(i) explain how ibuprofen reduces the effects of inflammation.

1. Ibuprofen is a (competitive) inhibitor

has a shape that can fit into the / enter the channel of access / 3D configuration complementary to active site;

2. Bind to the catalytic site, (form EI complex)

blocks AA from binding;

3. Prevent COX from converting AA

To prostaglandins;

[3]

(ii) explain how ibuprofen can bind to the same active site.

1. Ibuprofen may possess similar charges as AA

Form similar interactions as AA;

2. Can fold into the similar configuration as AA

To fit into the active site;

[2]

- © Inhibition of COX-1 by NSAIDs has been strongly implicated in gastric ulceration and bleeding. Newer drugs have been developed to selectively inhibit only COX-2.

Using information from Fig. 3.1 and 3.2, suggest how selective inhibition could occur.

Size of the drug to be bigger than the access channel for COX-1; [1]

[Total: 10]

- 4 To investigate the mode of DNA replication in a unicellular organism, 3 separate cultures of the cells were grown. The first was grown in a medium containing ^{14}N , the common isotope of nitrogen and the second was grown in a medium containing ^{15}N , the heavy isotope of nitrogen.

The cells of the third culture were grown in a medium containing ^{15}N for many generations first. Cells in early interphase were then transferred to a medium containing the common nitrogen form, ^{14}N . The cells were allowed to grow for one generation and then they were fixed at the next metaphase.

DNA was isolated from cells of all three cultures and separated into single strands by mild chemical treatment. The density of this single-stranded DNA was measured using density gradient centrifugation. The results of Tubes **A**, **B** and **C** are shown in Fig 3.1 below.

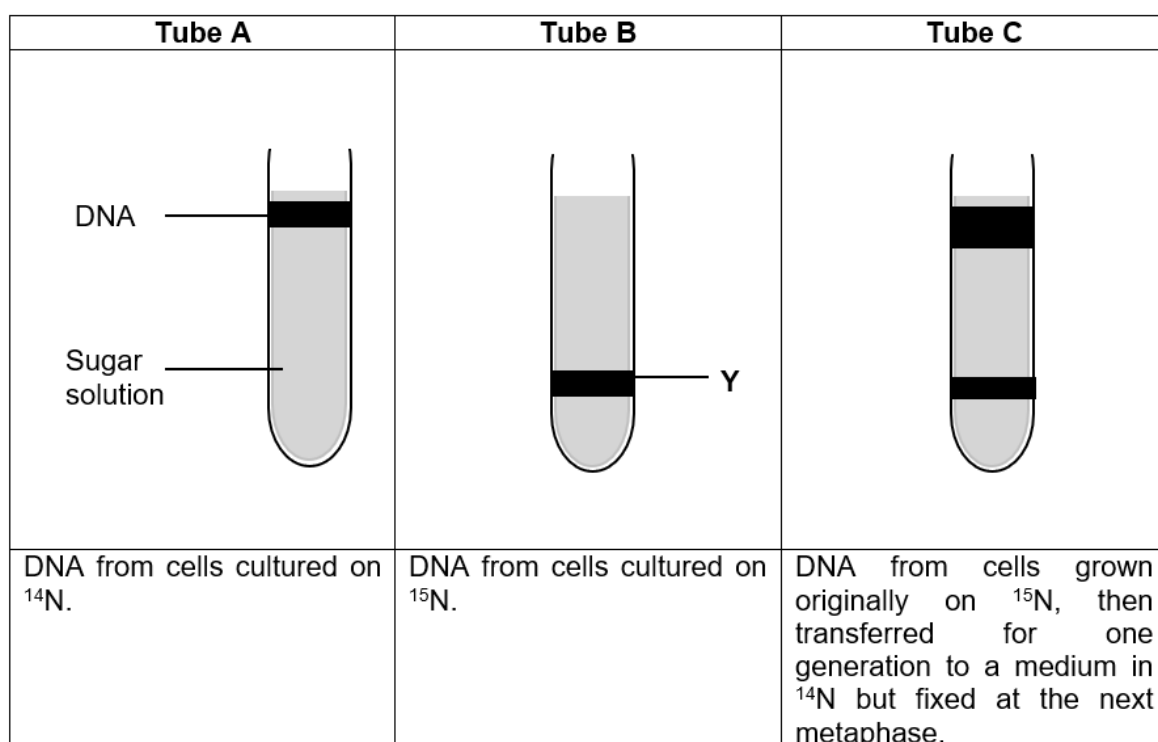


Fig. 4.1

- (a) (i) Assuming that DNA replication in the unicellular organism occurred through a similar mechanism to humans, what is the expected relative amount of each type of DNA in **Tube C**?

^{14}N : ^{15}N = 3:1 ratio / 75% : 25%

[1]

(ii) Explain your answer to (a)(i).

1. DNA replication is a semi-conservative process,

each of the original / parental strand acts as a template for the complementary daughter strand / newly synthesized strand;

2. at the end of first generation, the ratio of 1 ^{14}N : 1 ^{15}N DNA strands,

and after 2nd replication, ratio of 3 light: 1 heavy strand DNA or 3 ^{14}N : 1 ^{15}N DNA strands;;

OR

3. at the end of first generation, ^{15}N strands as template, to synthesize new complementary ^{14}N strand, resulting in ratio of 1 ^{14}N DNA: 1 ^{15}N DNA,

and after 2nd replication, both ^{15}N and ^{14}N strands are template for synthesis of new complementary ^{14}N strand,

there will be 3 light: 1 heavy strand DNA or 3 ^{14}N : 1 ^{15}N DNA strands;

[2]

Either 1 and 2 OR 1 and 3

Studies were then carried out to determine the nitrogenous base composition of the band of DNA marked Y in Tube B. This involved finding values for each individual (+ or -) DNA strand. Table 4.1 below shows the results obtained.

Table 4.1

DNA sample	Percentage of base present in DNA sample / %				Ratio of (A+G) to (C+T)
	A	G	C	T	
+ strand	28	15	25	32	0.75
- strand	32	25	15	28	1.33

(b) (i) Fill in the missing values in Table 4.1.

[1]

Correct values for C and T in all columns;

(ii) State the ratio of (A+G) to (C+T) in the band of DNA marked "Y" in Tube B in Fig. 4.1. Explain your answer.

1. Ratio of (A+G): (C+T) is 1:1;

2. Adenine and guanine are purines which will base pair respectively

with thymine and cytosine, which are pyrimidines;

[2]

- (iii) State why the ratios of (A+G) to (C+T) in the + and – strands of DNA do not follow the ratio stated in (b)(ii).

The + and – strands of DNA are single-stranded,

% of A, G, T and C is dependent on nucleotide sequence on the strand;

[1]

[Total: 7]

- 5 Occasionally during meiosis, homologous chromosomes fail to separate at anaphase. This is known as non-disjunction.

Turner's syndrome is the most common chromosome mutation in human females that occurs during gamete formation. Errors during meiosis led to some resulting gametes missing an X chromosome or when one of the pair of X chromosomes is damaged.

Fig. 5.1 is a diagram of a normal X chromosome and two variants of 'damaged' X chromosomes, X_1 and X_2 .

- In X_1 , a section of the 'p' arm of the chromosome is missing.
- In X_2 , a section of the 'q' arm of the chromosome is missing.

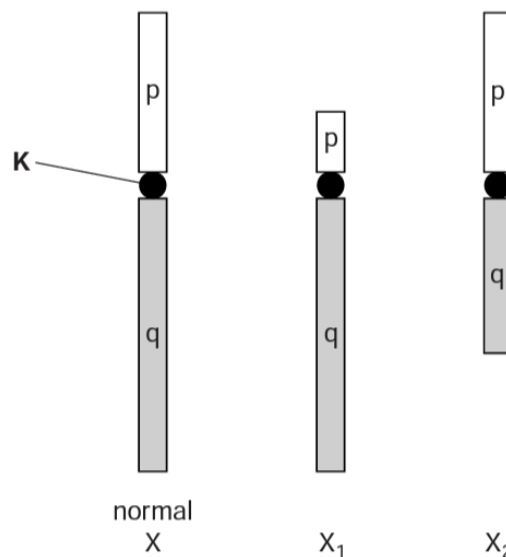


Fig. 5.1

- (a) State the function of structure **K**.

1. site of attachment of sister chromatids;

2. attachment site for kinetochore proteins;

[1]

any 1 MP

- (b) Suggest how variants X_1 and X_2 can be produced.

1. Cause of variant

e.g. spontaneous mutation / exposure to mutagen;

2. Impact on chromosome

e.g. translocation / breakage of chromosomes / deletion of chromosomal segments;

[2]

- (c) Explain how spindle fibres play an important role in ensuring a gamete with normal number of chromosomes is produced.

1. Kinetochore spindle fibres bind to all chromosomes

Allow spindle checkpoint to complete;

2. Ensure alignment of chromosomes

at metaphase plates during Metaphase (I and II);

3. Shorten to pull all chromosomes

To respective poles during Anaphase (I and II) for equal separation of chromosomes into daughter cells / gametes;

4. Non-kinetochore spindle fibres

Interact with each other to lengthen the cell;

[3]

any 3 marking points

- (d) The genes on the X chromosomes are involved in the development of the female reproductive organs.

Suggest why females with Turner's syndrome are sterile.

1. Inability to develop eggs, due to underdeveloped ovaries;

Inability to produce female hormones for development of womb lining / under-developed ovaries / uterus / inability to ovulate;

[1]

any 1 marking point

Unlike meiosis which occurs in the cells within reproductive organs, mitosis occurs in somatic cells found throughout the body.

At various points during the mitotic cell cycle, there are checkpoints to ensure that only genetically identical daughter cells to each other and to the parent cell are produced. Other cells which did not pass the checkpoints undergo apoptosis to prevent uncontrolled cell division.

Vincristine and 5-fluorouracil are drugs which have been developed to cause cellular apoptosis.

- Vincristine binds to spindle microtubules and prevents the spindle apparatus from carrying out its function.
 - 5-fluorouracil prevents the synthesis of thymine nucleotides
- (e) Complete Table 5.1 to show which event in the cell cycle will occur when Vincristine or 5-fluorouracil are added to healthy dividing cells at the start of the interphase stage of the cell cycle.

Place a tick (✓) if the event will occur or a cross (×) if the event will not occur. All boxes in the table should be completed.

Table 5.1

Cell cycle event Drug	Interphase	Prophase	Metaphase	Anaphase
Vincristine	✓	✓	×	×
5-fluorouracil	×	×	×	×

[2]

[Total: 9]

6 Fig. 6.1 shows the unpacking of bacterial circular chromosome.

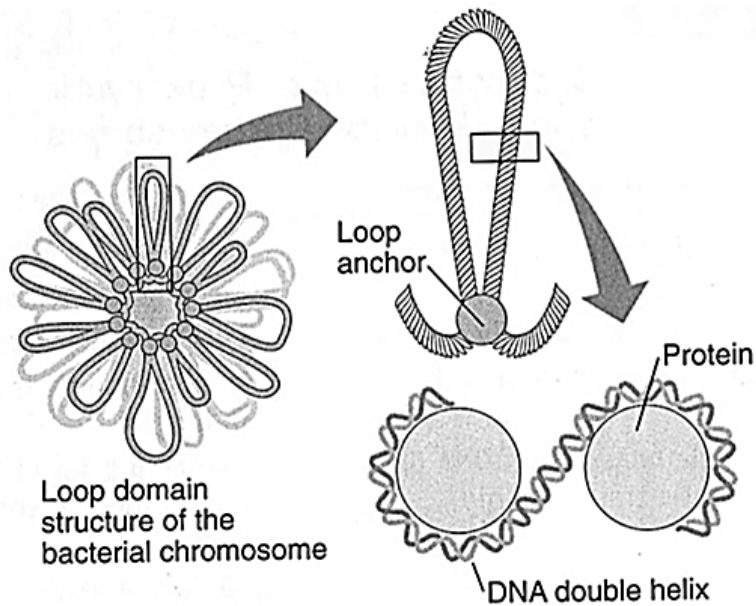


Fig. 6.1

(a) Using the information from Fig 6.1 and your own knowledge, describe how the bacterial chromosome is packed in a bacterial cell.

1. single molecule of circular DNA

With a double helix structure;

2. Associated / coil around (DNA binding) proteins

® DNA coils around loop anchor

to result in the DNA bending / looping;

3. Further coiling results / DNA held by loop anchor

in the formation of loop domains;

4. Supercoiling

Formed highly condensed chromosomes;

[4]

Fig. 6.2 shows the production of defective phages during bacteriophage replication that results in gene transfer between bacteria.

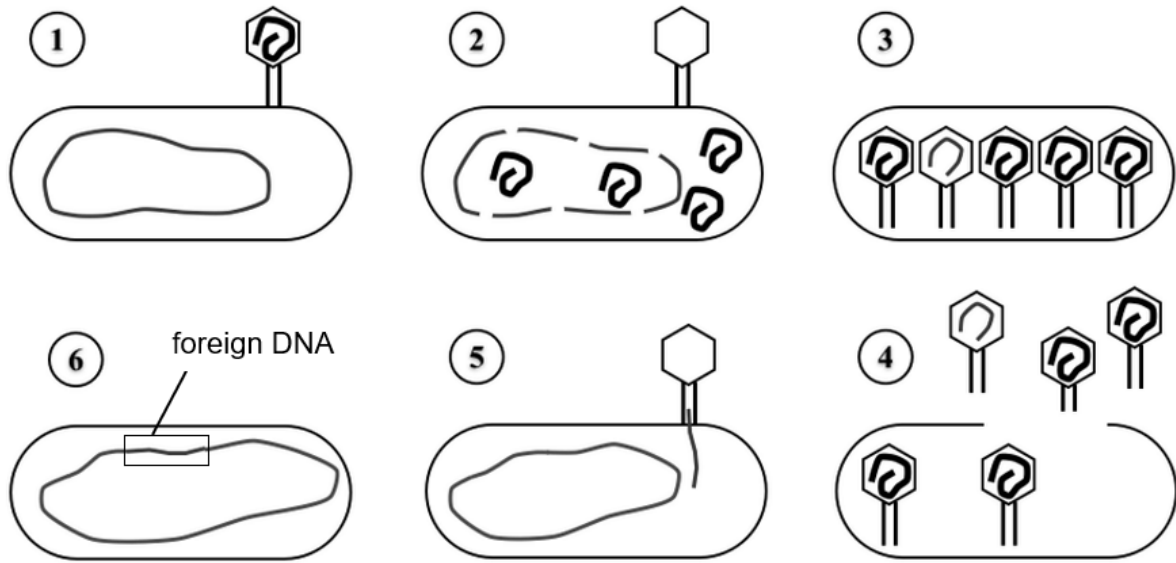


Fig. 6.2

(b) With reference to Fig. 6.2,

(i) suggest why Stage 2 occurs.

1. Prevent expression of bacteria genes

Coding for enzymes (@ phage-encoded nuclease)
(@: lysosome, lysozyme) that degrade bacteriophage DNA;

2. Frees up dNTPs

@ frees up DNA nucleotides

For replication of phage genome;

[2]

(ii) describe how the bacteria cells obtain new genetic material.

1. Random segments of bacteria DNA

packed into capsid head of bacteriophage;

2. Defective phage attaches to new host cell

with receptors on cell surface that are specific/ complementary to the tail fibres;

3. Bacterial DNA injected into new host cell,

homologous recombination occurred to result in integration of foreign DNA;

[3]

[Total: 9]

- 7 The interpupillary distance (IPD) is the distance in millimetres between the centres of the pupils of the eyes. Fig. 7.1 shows how IPD is measured.

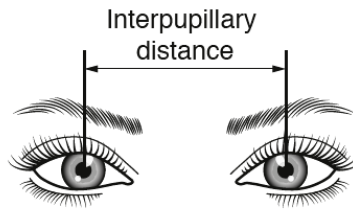


Fig. 7.1

IPD is one example of a characteristic of human facial structure that shows variation. Fig. 7.2 shows the pattern of variation in IPD in a large sample of adults.

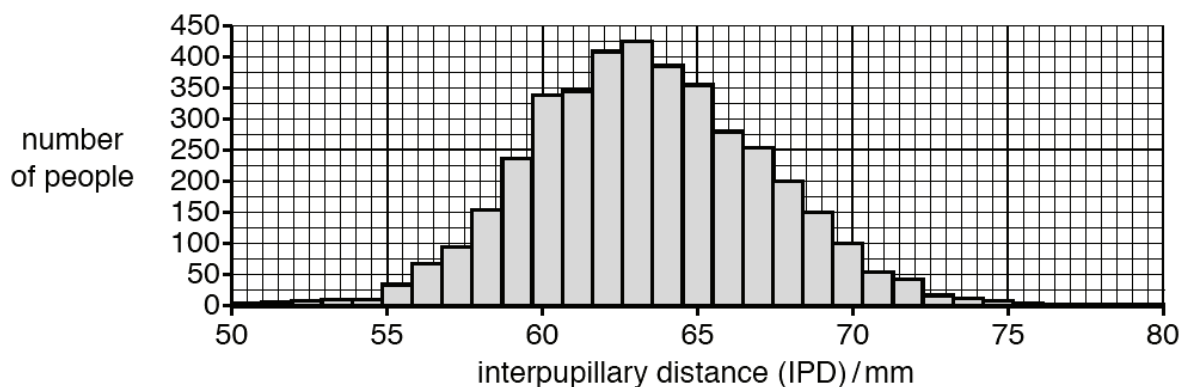


Fig. 7.2

- (a) (i) Name the type of variation shown in Fig. 7.2.
continuous variation; [1]
- (ii) Explain how genes and the environment contribute to variation in IPD in humans.
 1. **IPD is controlled by several / many genes / polygenic trait;**
 2. **additive effect of genes;**
 3. **each gene has small influence / impact;**
 4. **environmental factors affect gene expression;**
named e.g. diet / age; [3]

Max. 2m from marking points 1 to 3

An investigation was carried out on the difference in IPD of males and females. Table 7.1 shows the results of the investigation.

Table 7.1

individual	IPD / mm	
	male	female
1	70	65
2	72	62
3	68	59
4	73	69
5	72	66
6	65	63
7	66	67
8	62	58
9	67	62
10	68	68
mean ($\bar{x}$)	68	64
standard deviation (s)	3.50	3.73
variance (s^2)	12.25	13.91

(A) 3 sig. fig

- (b) (i) Complete Table 7.1 by calculating the variance (s^2) for the IPD for each gender. [1]

- (ii) A t-test can be used to determine whether there is any significant difference between the IPD in males and females.

Calculate the value of t and the number of degrees of freedom, using these formulae:

$$t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}} \quad v = n_1 + n_2 - 2$$

key to symbols

s = standard deviation

$\bar{x}$ = mean

n = sample size (number of observations)

v = degrees of freedom

Show your working.

$$t = \frac{|68 - 64|}{\sqrt{\frac{12.25}{10} + \frac{13.9}{10}}}$$

$$= 2.47$$

value of t = 2.47;

@ 2 dp / 3 sf

@ variation from
rounding off

degree of freedom = 18; [2]

(iii) For this t-test, the proposed null hypothesis is:

there is no difference between the IPD between males and females.

Table 7.2 shows the critical values for several different probabilities and degrees of freedom.

Table 7.2

degrees of freedom	probability, p				
	0.5	0.1	0.05	0.01	0.001
1	1.00	6.31	12.71	63.66	636.62
2	0.82	2.92	4.30	9.92	31.60
3	0.76	2.35	3.18	5.84	12.92
4	0.74	2.13	2.78	4.60	8.61
5	0.73	2.02	2.57	4.03	6.87
6	0.72	1.94	2.45	3.71	5.96
7	0.71	1.89	2.36	3.50	5.41
8	0.71	1.86	2.31	3.36	5.04
9	0.70	1.83	2.26	3.25	4.78
10	0.70	1.81	2.23	3.17	4.59
11	0.70	1.80	2.20	3.11	4.44
12	0.70	1.78	2.18	3.05	4.32
13	0.69	1.77	2.16	3.01	4.22
14	0.69	1.76	2.14	2.98	4.14
15	0.69	1.75	2.13	2.95	4.07
16	0.69	1.75	2.12	2.92	4.01
17	0.69	1.74	2.11	2.90	3.97
18	0.69	1.73	2.10	2.88	3.92
19	0.69	1.73	2.09	2.86	3.88
20	0.69	1.72	2.09	2.85	3.85

Use Table 7.2 and your answers to **(b)(ii)** to decide whether the null hypothesis should be accepted or rejected.

Explain your answer.

accept or reject null hypothesis **Reject;**

explanation **1. calculated t value of 2.47 is more than critical t value of 2.10**

(at $v = 18$, $p = 0.05$);

2. difference in IPD between male and female IPD is statistically significant

difference is not due to chance;

[2]

[Total: 9]

8 Photosynthesis is a complex process involving a light-dependent stage and a light-independent stage.

- (a) (i) Name the products of the light-dependent stage that are needed in the light-independent stage.

ATP, NADPH / reduced NADP;

[1]

- (ii) Describe the role of chlorophyll b in photosynthesis.

act as accessory pigment (in light harvesting complex)

absorbs light to pass on to reaction centre / chlorophyll a molecule (via inductive resonance);

[1]

A student carried out an experiment to investigate the effect of light intensity and light wavelength on the rate of photosynthesis.

An aquatic plant, *Elodea canadensis*, was put into a beaker containing sodium hydrogencarbonate solution as a source of carbon dioxide. A lamp was placed at different distances from the beaker to change light intensity. The number of bubbles released by the aquatic plant in 1 minute was counted as a measure of the rate of photosynthesis. All other variables were controlled.

At each distance from the lamp, the experiment was repeated using a red filter in front of the lamp to give a different wavelength of light. The experiment was repeated using a blue filter and then using a green filter. Each filter transmitted the same light intensity.

Fig. 8.1 shows a graph of the results.

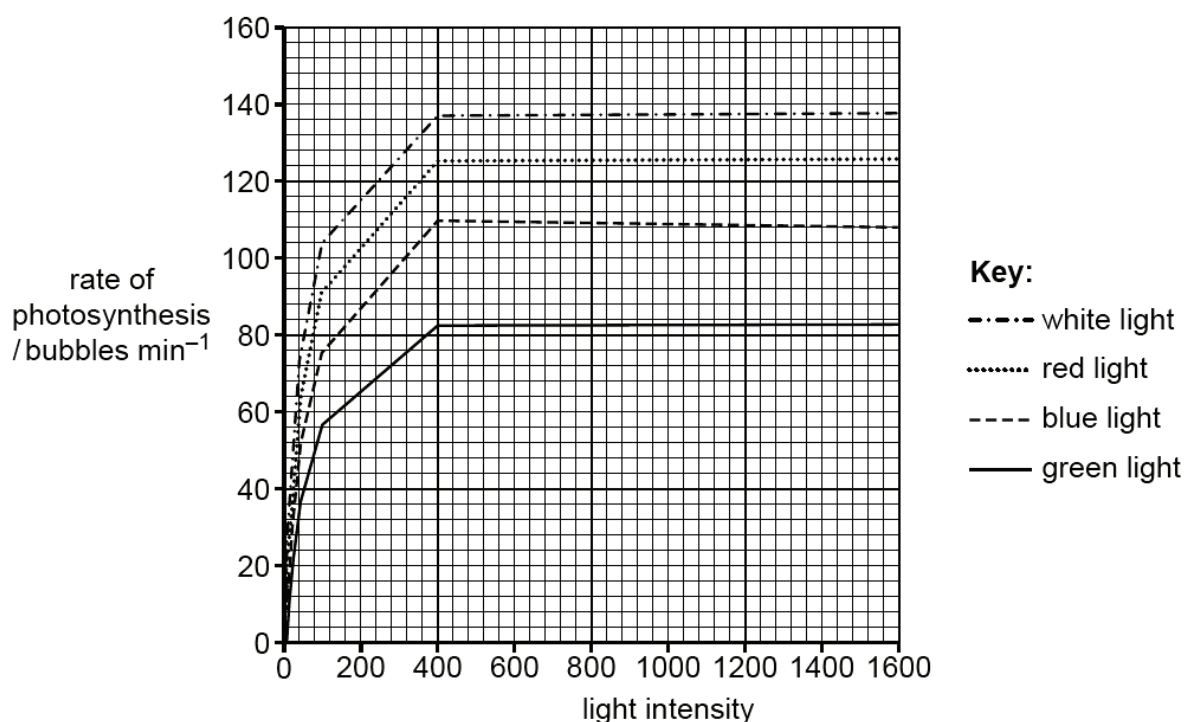


Fig. 8.1

(b) With reference to Fig. 8.1,

- (i) state the range over which light intensity is the limiting factor;

0 – 400;

[1]

- (ii) explain what factor may be limiting the rate of photosynthesis above the range of light intensity stated in (c)(i).

1. temperature;

2. less kinetic energy

result in lower rate of (named enzyme-controlled reaction) e.g. effective collisions between H_2O and water-splitting complex;

3. concentration of CO_2

4. results in less frequent effective collision between CO_2 and rubisco thus less CO_2 fixation in Calvin cycle;

[2]

Marking points 1+2 or 3+4

- (iii) At a light intensity of 1600, explain why different colour filters result in different rates of photosynthesis.

1. rate of photosynthesis is highest for red light at 125 bubbles min^{-1}

lowest for green light at 83/84 bubbles min^{-1} ;

2. different wavelengths of light

is absorbed by different pigments (in photosystem);

3. red light is absorbed the most

leading to faster rate of light dependent reaction / photophosphorylation / photoactivation (of chlorophyll a) / ORA;

[3]

- (iv) Predict and explain the effect on the number of bubbles produced if a rubisco inhibitor was added.

1. number of bubbles produced would decrease;

2. inhibitor lowers rate of CO_2 fixation by rubisco

thus lower rate of Calvin cycle;

3. lower regeneration of ADP and NADP

leading to lower rate of light dependent reactions thus less O_2 released from photolysis;

[3]

[Total: 11]

- 9 The aye-aye, *Daubentonia madagascariensis*, is a primate native to Madagascar. Aye-ayes are nocturnal (active at night) and make their nests high up in trees. They feed on insect larvae in the trunks of trees.

Fig. 9.1 shows an aye-aye.



Fig. 9.1

The International Union for Conservation of Nature (IUCN) is the world's largest global environmental organisation. The IUCN Red List of Threatened Species™ evaluates the conservation status of plant and animal species.

The aye-aye is categorised as endangered on the IUCN Red List, which means that it faces a very high risk of becoming extinct in the wild.

- (a) (i) Table 9.1 shows the taxonomic classification of aye-aye.

Table 9.1

Domain	Eukarya
Kingdom	Animalia
Phylum	Chordata
Class	Mammalia
Order	Primates
Family	Daubentoniidae
Genus	<u><i>Daubentonia</i></u>
Species	<u><i>Daubentonia madagascariensis</i></u>

Complete Table 9.1.

[3]

Marking points:

1. Taxonomy: all correct for credit;
2. Correct Genus name;
3. Correct species name, underlined;

- (ii) Suggest **one** reason why aye-ayes have become endangered.

habitat destruction / hunting / AVP with brief elaboration;

[1]

There are two main aye-aye populations on the island of Madagascar, one in the west and one in the east.

Fig. 9.2 is a map of Madagascar showing the location of the two main populations.

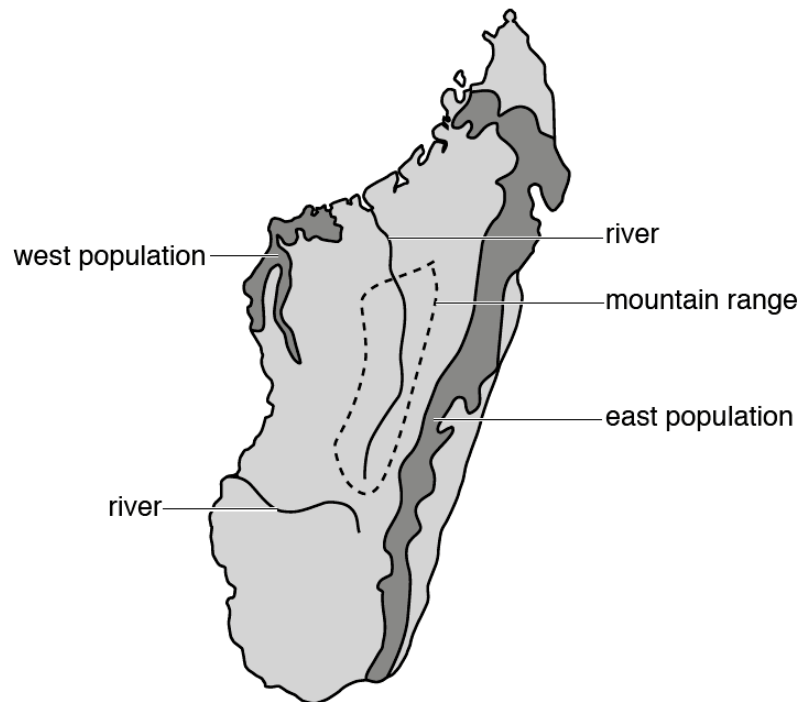


Fig. 9.2

A study into the variation in the DNA nucleotide sequence of aye-ayes showed that there is a large genetic difference between the west and east populations. The two populations of aye-ayes may be evolving into separate species.

- (b) (i) State one advantage of using DNA nucleotide sequence in this study.

unambiguous / objective / quantifiable / AVP;

[1]

- (ii) With reference to Fig. 9.2, suggest why there is a large genetic difference between the two populations.

1. east and west populations of aye-ayes have phenotypic variation

due to presence of genotypic variation;

2. east and west of Madagascar have different environmental conditions thus different selection pressures

resulting in aye-ayes with different favourable phenotypes being selected for;

3. different favourable alleles are passed down to offspring

leading to different allelic frequencies in the gene pools;

4. different mutations occur in the east and west populations;

5. geographical isolation / barrier e.g. rivers / mountains between the 2 populations

prevent interbreeding / gene flow resulting in genetic divergence; [4]

Any 4 marking points

(iii) Name the type of speciation that may be occurring.

allopatric speciation;

[1]

(iv) Suggest **and** explain a pre-zygotic isolating mechanism that could prevent successful reproduction between aye-ayes of the two populations.

1. physical / morphological / mechanical;

2. reproductive features do not match / unable to mate;

3. behavioural;

4. different mating calls / courtship rituals;

5. gametic;

6. incompatible gametes causing fertilisation to be unsuccessful;

7. temporal;

8. different breeding seasons;

[2]

Marking points 1+2 / 3+4 / 5+6 / 7+8

[Total: 12]

10 Tuberculosis (TB) is a major cause of ill health worldwide.

- (a) The World Health Organization (WHO) Global Tuberculosis Report for 2019 published data on the estimated number of deaths from TB and HIV/AIDS in 2018. All deaths of HIV-infected people from TB were also counted as deaths of people with HIV/AIDS.

Fig. 10.1 shows these data. The dark grey boxes show the estimated number of deaths of people from TB who were also counted as deaths of people with HIV/AIDS.

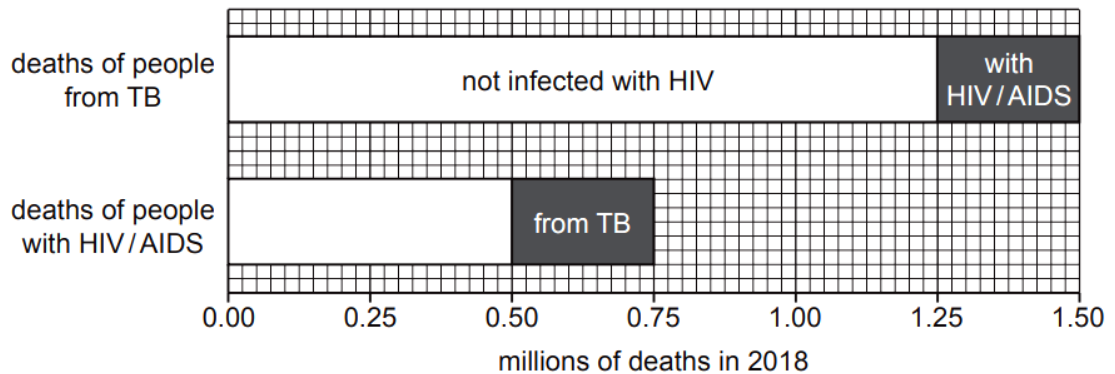


Fig. 10.1

A student used the data in Fig. 10.1 to predict that measures to control the spread of HIV will decrease the number of deaths from TB.

Discuss whether the data in Fig. 10.1 support this prediction.

[Does not support]

1. much larger proportion of individuals dying from TB do not have HIV/AIDS;
2. [QV] deaths of people from TB with HIV/AIDS is 0.25 million out of 1.5 million
- OR 1.25 million people dying from TB does not have HIV/AIDS;
3. the 0.25 million people may still die from TB even if they do not have HIV/AIDS;

[Support]

4. significant proportion of deaths of people from with HIV/AIDS are caused by TB;
5. [QV] 0.25 million deaths of people with HIV / AIDS caused by TB out of 0.75 million;
6. fewer people will be immunocompromised thus less susceptible to TB;
7. AVP;

[3]

- (b) In healthy people, the number of T-helper cells ranges from 500 to 1200 cells per cm^3 of blood. In untreated people infected with HIV, the number of T-helper cells can decrease to below 200 cells per cm^3 of blood.

Explain how a low number of T-helper cells makes it more likely that untreated people infected with HIV will die if they are also infected with TB.

1. lesser activation of B cells

to undergo clonal expansion and differentiation (to form effector B cells);

2. fewer plasma cells to produce / secrete antibodies;

3. fewer memory B cells to fight future infections;

4. lesser activation of CD8+ cells

to undergo clonal expansion and differentiation (to form effector T cells);

5. fewer cytotoxic T cells to kill infected cells;

6. fewer memory T cells to fight future infections;

7. people infected with HIV is immunocompromised

increased risk of developing active TB / succumb to opportunistic infections;

[4]

[Total: 7]

- 11 Frogs are ectothermic animals. This means that their body temperature will vary as the environmental temperature varies.

Several species of the frog genus, *Rana*, can be found in North America. Many of these species inhabit areas within a range of latitudes from the colder north to the warmer south.

Table 11.1 shows data for four of these species, *R. clamitans*, *R. palustris*, *R. pipens* and *R. sylvatica*.

Table 11.1

Species	Body temperature of frog / °C			
	lower lethal, below which frog dies	minimum to start development	maximum to complete development	upper lethal, above which frog dies
<i>R. clamitans</i>	10.0	11.0	35.0	37.0
<i>R. palustris</i>	5.0	7.0	30.0	31.0
<i>R. pipens</i>	3.0	6.0	28.0	30.0
<i>R. sylvatica</i>	0.0	2.0	24.0	25.0

- (a) Using the information in Table 11.1, suggest why global warming will result in a decrease in species diversity in frogs in the current habitats.

1. frogs are ectotherms / body temperature varies with environmental temperature

global warming will increase frogs' body temperature;

2. different frog species have different upper lethal range

25°C in R. sylvatica to 37°C in R. clamitans;

3. species from habitats at lower latitude (e.g. R. clamitans) may migrate to higher latitudes / towards the north

as habitat temperatures exceeds upper lethal temperature;

4. species already from habitats at higher latitudes (e.g. R. sylvatica) are not able to migrate further north

thus die when environmental temperature increases above upper lethal temperature;

[3]

any 3 Marking points

- (b) Explain how human activities could have contributed to an increase in greenhouse gases that resulted in the rise of global temperatures.

1. state any one anthropogenic effect that contributed to CO₂

e.g. burning of fossil fuel due to transportation, powering of buildings, etc.;

2. state any one anthropogenic effect that contributed to CH₄

e.g. cattle rearing due to increased consumption of meat;

[2]

[Total: 5]